

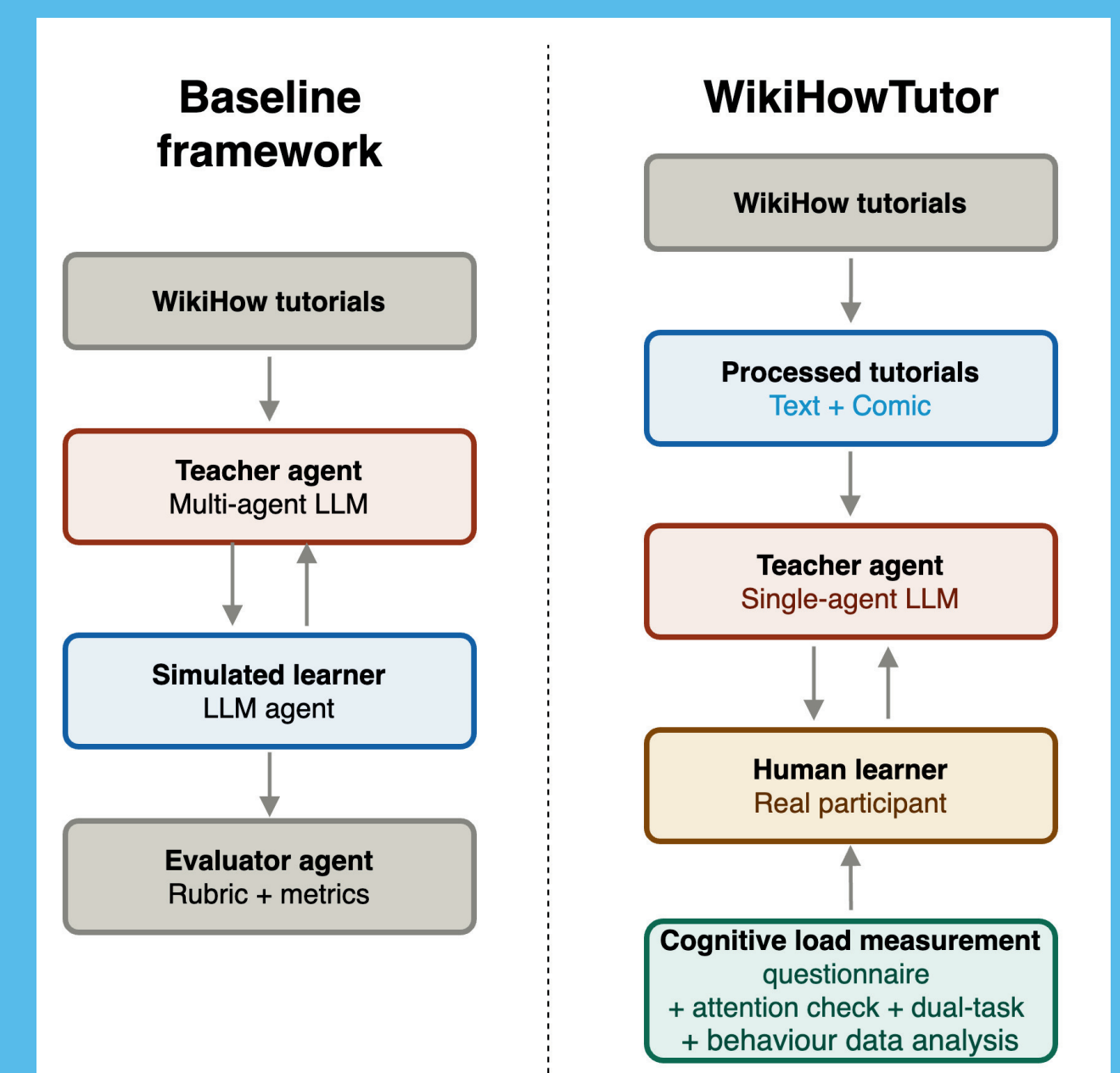
Measuring Human Cognitive Load in AI-Assisted Tutorial Learning: A Comparison of Text and Comic Materials

Author by: Jiaqi Shi

Supervised by: dr. Jiahuan Pei, Yue Su



We compare how **text and comic format** affects human **cognitive load** in AI-assisted tutorial learning with **WikiHowTutor**, and find that comic significantly reduces cognitive load without compromising learning.



The Problem

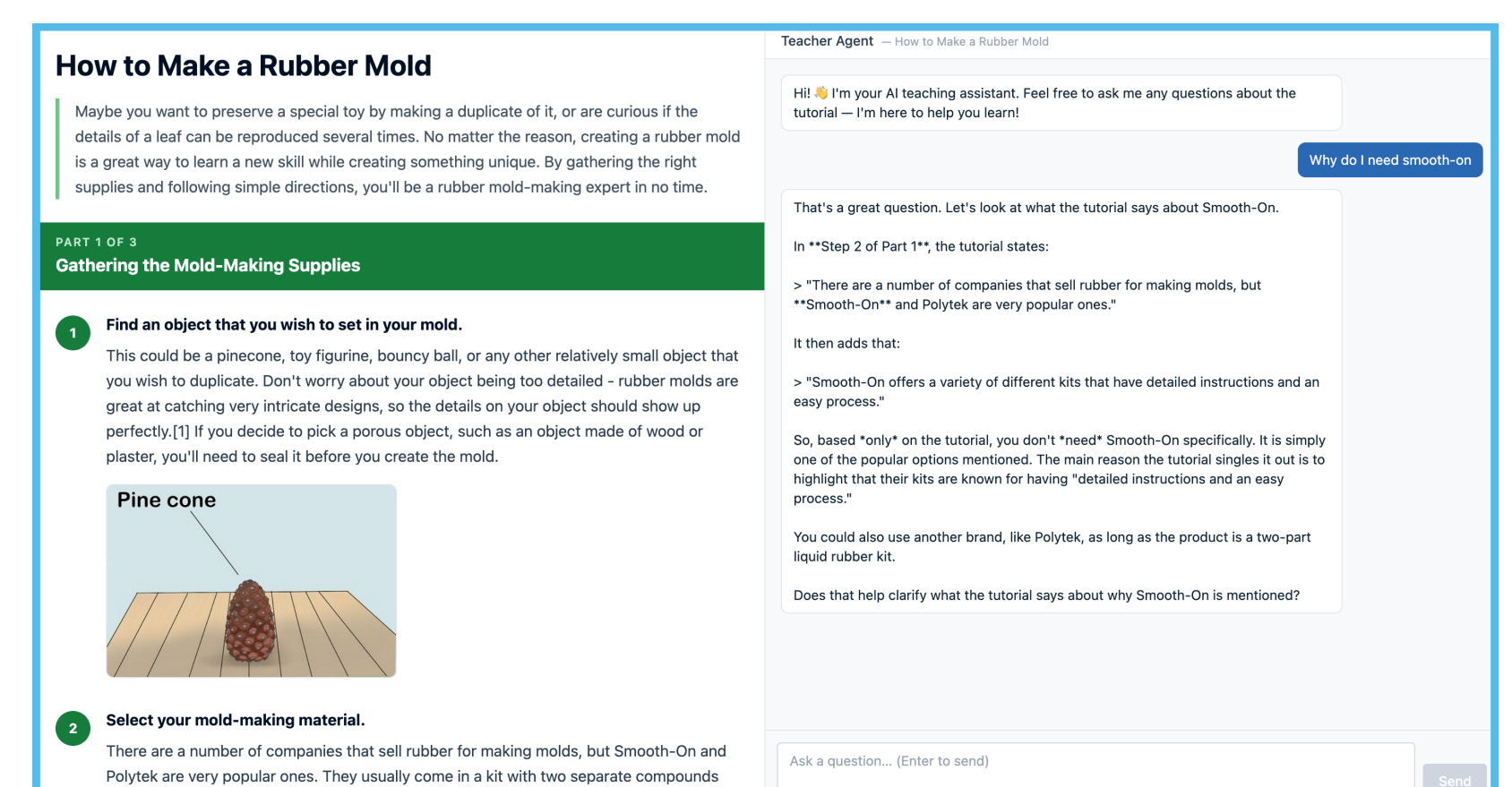
- AI tutoring research focuses more on agent performance than learner experience.
- Material format affects cognitive load, but existing studies lack AI-assisted contexts
- WikiHowAgent (basework) used simulated learners, not human learners.

Research Questions

- How does material format affect cognitive load for human learners?
- Can adjusting cognitive load enhance the learner experience?
- How can cognitive load be detected in real-time?

Our Approach : WikihowTutor

- We built an LLM-based tutoring platform, selected, paired, and formatted 15 pairs of WikiHow tutorials, to measure and analyze user's cognitive load.
- Procedure: Consent → pre-survey → session A → post-survey → session B → post-survey
- Measures: CL questionnaire (ICL/ECL/GCL) + UMUX-Lite + quiz + behavioral signals + dual-task probe



User Interface Example (More in Demo)

Result

RQ1 (Format - Cognitive Load)

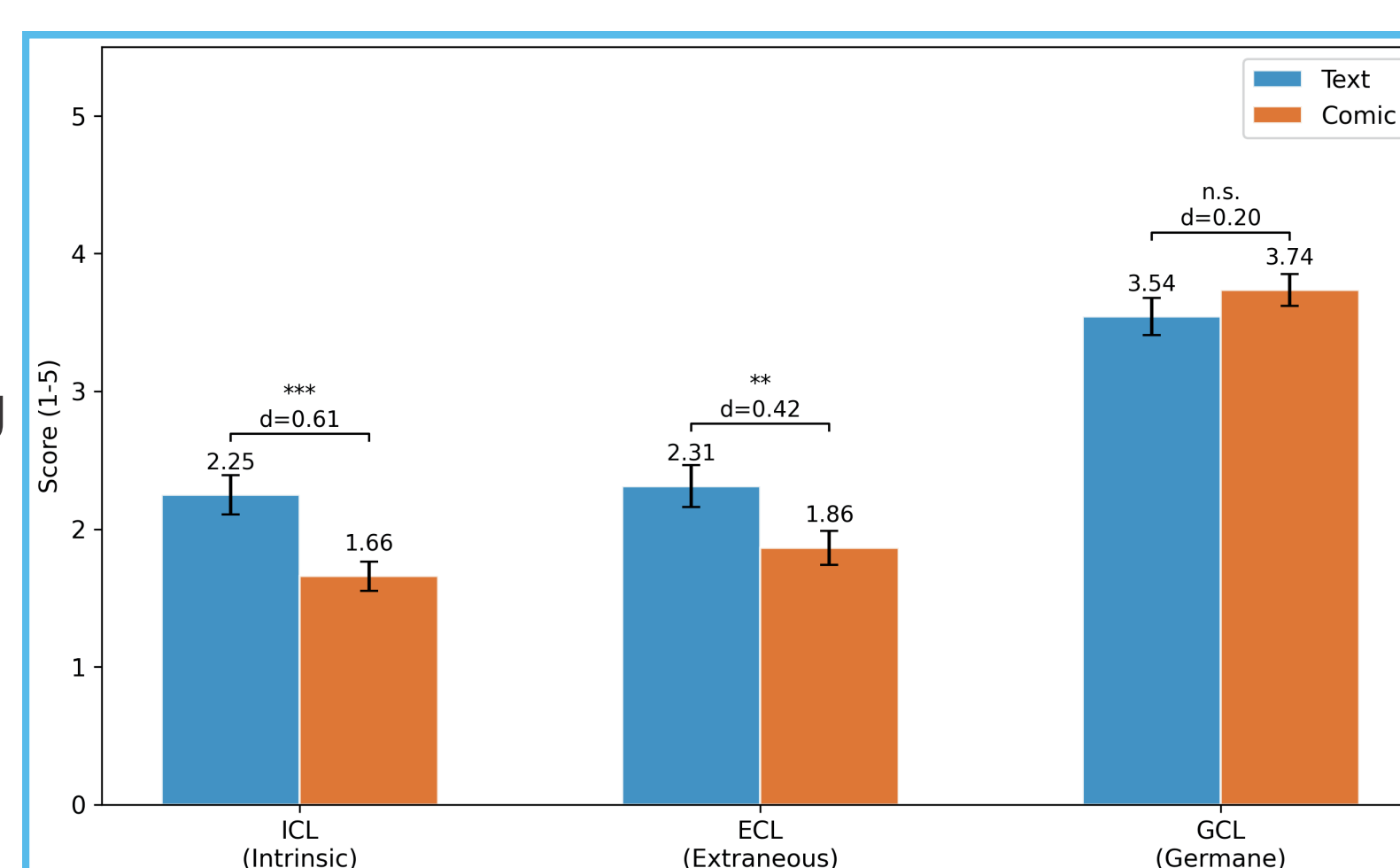
- MANOVA significant: $F(3,56)=4.98$, $p=.004$
- ICL & ECL: text > comic ($d=0.61$)
- GCL: no significant difference

→ **Comic reduces CL without harming learning**

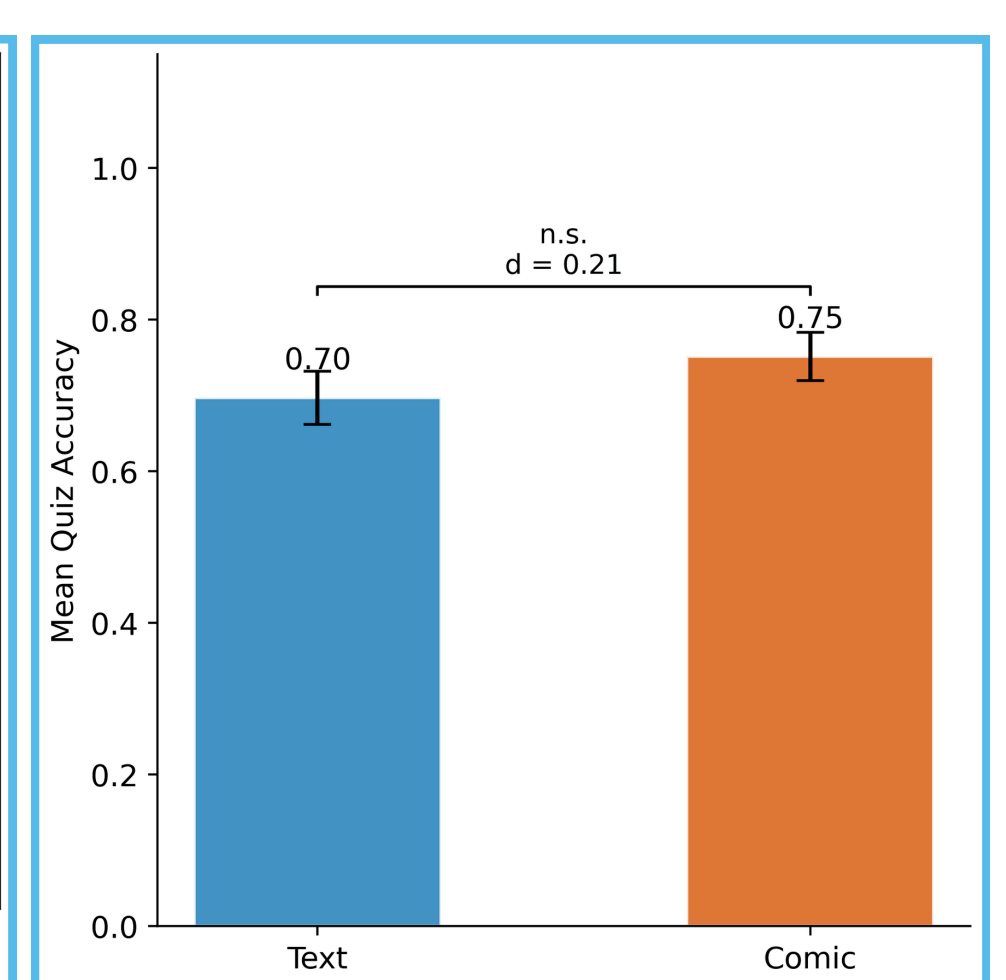
RQ2 (Cognitive Load - Learner Experience)

- Text: higher ICL/ECL → lower usability
- Comic: higher GCL → higher usability
- Format change in CL predicts usability change ($\Delta\text{ICL}-\Delta\text{UMUX}$: $r=-.31$, $p=.018$)

→ **Adjusting CL through format can affect learner experience**



Mean Cognitive Load Scores by Format



Mean Quiz Accuracy by Format

Summary & Limit

- Comic format leads to significantly lower ICL and ECL than text, with no significant difference in learning outcomes.
- Cognitive load drives learner experience differently: ECL and ICL lowers it in text, while GCL improves it in comic.
- Real-time cognitive load detection was limited by insufficient dual-task data and small per-condition sample size.